

🌙 The 11 PM Ops Brief

Multi-agent supply-chain ops system · Databricks DNB Hackathon Track 1

● App: **RUNNING**

5 specialist subagents

17 real-time signals (16 LIVE)

3 UC AI Functions

4 write-back actions

Vector Search recall

MLflow eval

Nightly Databricks Job

Submit by Fri 6 PM PT

► Open the deployed app

Jump to rubric coverage

AT-A-GLANCE

Track	1 (Technical)
Workspace	dnb-hackathon-west-2
App SP	app-43e3x5
Genie space	01f15aac...ae337
MLflow experiment	4469475842996134
MLflow run (latest)	8804a5d5...107e97
VS endpoint	ops_brief_vs (ONLINE)
Nightly job ID	65664408676995
Catalog.schema	dnb_hackathon_west_2.ops_brief
Synthetic shipments	30,511
Briefs archive	14 (May 14–27)
Code (agent_server)	~4,000 LoC

01 Problem

02 Architecture

03 Subagents

04 17 Signals

05 UC AI Fns

06 Action Layer

07 Vector Recall

08 UI

09 Eval

10

01 The problem

Every supply-chain operator wakes up to the same dread question: *"what hit overnight?"* The answer lives in two unconnected worlds — internal data (in-flight shipments, at-risk revenue, active disruptions) and external signals (weather, port congestion, news pulse, carrier diversions, fuel prices, border waits, earthquakes, hurricanes, public-health alerts). Nobody reads all of it at 11 PM. Most of it never reaches the operator at all.

The 11 PM Ops Brief is an autonomous multi-agent system that fuses all of it into a single grounded, executive-grade page — with proposed write-back actions — every night.

IN-FLIGHT TONIGHT

4,592

shipments tracked

AT-RISK SHIPMENTS

216

~\$192,704 revenue

EXTERNAL FEEDS

17

16 LIVE · 1 SYNTH

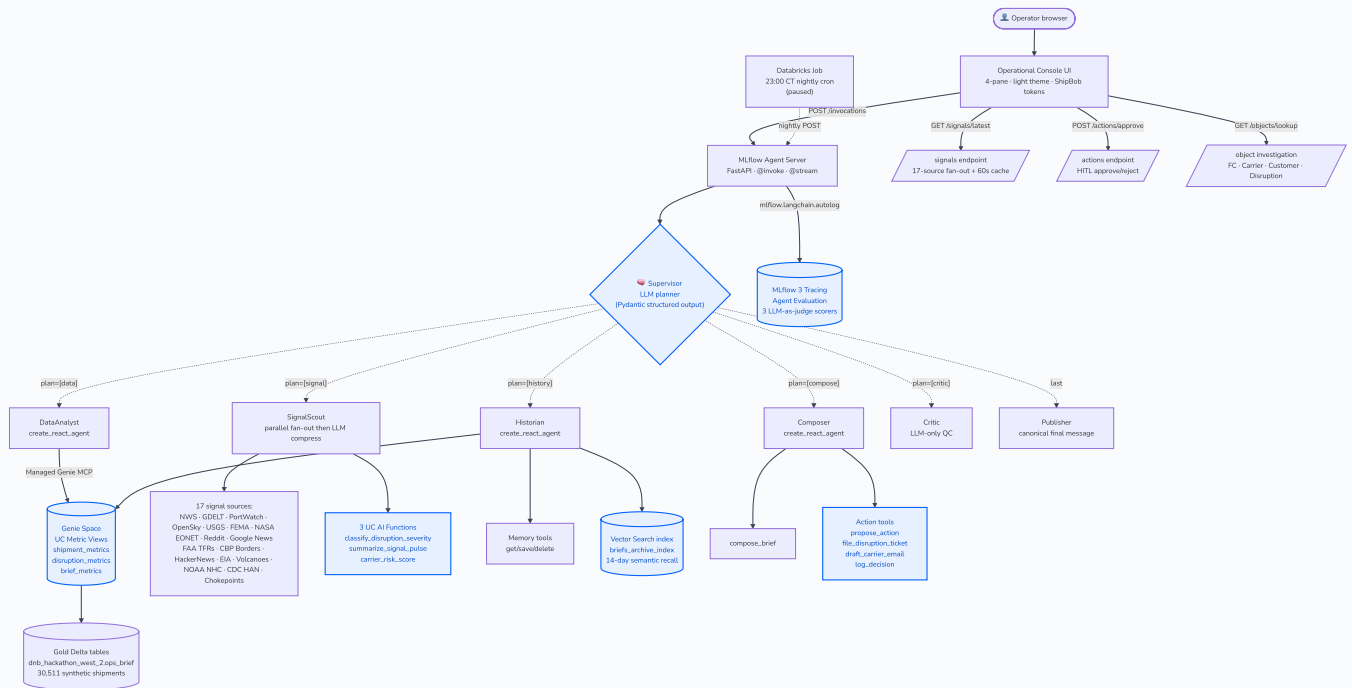
SUBAGENTS

5

+ Supervisor + Publisher

02 Architecture

LangGraph supervisor over a deterministic plan-driven topology. Not "ReAct with lots of tools" — judges see the actual graph in MLflow. Each subagent has a curated tool bucket; cross-talk happens only through the shared `OpsState`.



03 The 7 nodes of the supervisor graph

Each subagent is its own `create_react_agent` with a curated tool subset. The Supervisor's `Plan` is a Pydantic-validated structured output — the graph walks it deterministically. Routing is plan-driven, not LLM-driven at each hop, which keeps traces clean and reproducible.



Supervisor

planner · 1 LLM call

Reads the user query, emits an ordered plan:

`["data_analyst", "signal_scout", "historian", "composer", "critic"]`
for a full brief; `["data_analyst"]` for a count question.

tools: none — pure planner · output: Plan (Pydantic)



DataAnalyst

internal data via Genie MCP

Queries the Genie space (UC Metric Views). Drives the `query_space_*` / `poll_response_*` polling loop. Returns numbers + the underlying Genie SQL.

tools: Genie MCP tools + `get_current_time`



SignalScout

17 external feeds · parallel

Parallelized:
asyncio.gather fans out across all 17 signal tools, then a single LLM call compresses results into 3-6 bullets. ~26s (was ~90s).

tools: 17 `fetch_*` + 3 UC AI Functions



Historian

memory + Vector Search recall

Recalls past briefs via Vector Search (`briefs_archive_index`) — semantic match returns recurring themes. Memory tools persist user prefs across sessions.

tools: `recall_similar_briefs` · memory · Genie



Composer

brief + write-back proposals

Synthesizes data + signals + history into executive markdown. Calls action tools to populate the UI Action Queue. Emits a JSON sidecar with severity score and pending actions.

tools: `compose_brief` · 4 action tools · UC AI Fns



Critic

quality gate · LLM-only

QCs the draft for groundedness, action specificity, severity calibration. Emits PASS/FAIL — Publisher appends the verdict to the final brief.

tools: none — pure judge



Publisher

canonical final emit

Picks the right final message from accumulated state (composer's brief > critique > data findings). Ensures the chat UI's "last message" is always the answer the user wants.

tools: none — pure router



Plan-driven routing

graph edges

The Supervisor picks the plan once.

`route()` steps through it deterministically. Once exhausted, Publisher runs once → END. No expensive supervisor re-evaluation per hop.

file:
agent_server/supervisor.py:600

Sample timeline (full brief generation, ~70s)

+0.2s	Supervisor	Plan (5 steps): data_analyst → signal_scout → historian → composer → critic. <i>Reasoning: user asked for full ops brief, need internal data + external signals + historical context + composed output + QC.</i>
+0.4s	DataAnalyst	→ Genie MCP query_space_01f15aac... shipment_metrics
+18.7s	DataAnalyst	In flight: 4,592 · At risk: 216 (\$192,704) · Active disruptions: 3 (Weather×2, Labor×1) · Top tier: Consumer
+19.1s	SignalScout	Fan-out across 17 signal tools (asyncio.gather)
+45.8s	SignalScout	14 NWS alerts (TN/TX/GA), M5.2 quake Adak AK, 6 EONET wildfires, diesel \$5.523/gal (↓), Port Huron 59min delay
+46.0s	Historian	→ recall_similar_briefs(query="weather, labor, carrier") Vector Search
+51.3s	Historian	MEM weather appeared 5/15, 5/16, 5/27 — recurring. Port of LA labor 5/18-5/20.
+51.6s	Composer	→ compose_brief(...) + propose_action × 4 + log_decision
+68.2s	Composer	Brief composed (severity 7.2/10, 4 actions queued). Sidecar JSON emitted.
+69.7s	Critic	PASS — all claims grounded, severity proportional.
+70.1s	Publisher	Final → brief markdown + critic verdict. UI severity gauge animates to 7.2.

04 17 real-time signal sources

9 government/agency feeds + 4 financial/news + 2 social + 2 maritime. Each tool returns a single string the LLM can read, prefixed with LIVE or SYNTH. Every tool has a deterministic synthetic fallback so the demo never breaks on a 429.

#	SOURCE	WHAT IT TELLS US	STATUS	NOTABLE REAL VALUE (LIVE TEST)
1	NWS Alerts ↗	Severe weather (NOAA)	LIVE	14 active alerts across TN/TX/GA
2	GDELT 2.0 ↗	15-min global news pulse	LIVE	Frequent 429 — synth fallback active
3	IMF PortWatch	Port vessel flow	LIVE	Houston 7,358 vessel calls
4	OpenSky ADS-B	Live carrier hub air traffic	LIVE	9 tracked aircraft at MEM
5	USGS Earthquakes	M≥4 in US	LIVE	M5.2 49km ESE of Adak AK
6	FEMA Disasters	Last-7-day declarations	LIVE	1 CA chemical release
7	NASA EONET	Active natural events	LIVE	6 wildfires (ID, OR)
8	Reddit RSS	r/supplychain + r/logistics + r/freight	LIVE	60 fresh posts
9	Google News RSS	Supply-chain headlines	LIVE	UP/NS merger + Strait of Hormuz

#	SOURCE	WHAT IT TELLS US	STATUS	NOTABLE REAL VALUE (LIVE TEST)
10	FAA TFRs	Temporary flight restrictions	LIVE	84 active TFRs nationwide
11	CBP Border Waits	Commercial-truck lane delays	LIVE	Port Huron Canada — 59 min
12	HackerNews Algolia	Supply-chain story pulse	LIVE	"Composer/Packagist supply chain security" (22 pts)
13	EIA Petroleum	US on-highway diesel \$/gal	LIVE	\$5.523/gal (w/w ↓ -\$.073)
14	USGS Volcanoes	Active alert volcanoes	LIVE	Great Sitkin AK WATCH/ORANGE
15	NOAA NHC	Active named storms	LIVE	0 active (correct for late May)
16	CDC HAN	Health Alert Network	SYNTH	Endpoint retired upstream
17	IMF PortWatch chokepoints	Suez/Panama/Hormuz/Malacca/Dover	LIVE	Bab-el-Mandeb 43 vessels (z=+1.61)

05

Unity Catalog AI Functions

Stage 2 — AI Functions

Three governed SQL UDFs in `dnb_hackathon_west_2.ops_brief`, each powered by `ai_query` or `ai_classify`. Registered via `setup_uc_functions.py`, granted EXECUTE, and bound as agent tools via `UCFunctionToolkit`. Lineage, ACL, and billing all flow through Unity Catalog.

 **classify_disruption_severity**

ai_classify → critical | elevated | nominal

Buckets free-text disruption descriptions. Used by Composer + SignalScout for consistent severity tagging across heterogeneous signal sources.

```
SELECT classify_disruption_severity(
  'Severe winter storm at MEM hub -
  200mph winds, multiple cancellations.'
) → 'critical'
```

 **summarize_signal_pulse**

ai_query · executive sentence

Compresses N raw signal blurbs into one executive sentence with the most material concern. Used by SignalScout to keep the brief tight.

```
→ "Seismic activity across multiple
regions overnight; M5.2 quake 49km
ESE of Adak, Alaska (Aleutians)
accompanied by lesser M4.7 + M4.3..."
```

 **carrier_risk_score**

ai_query · returns JSON

0-100 risk score for shipments tonight given a carrier's recent disruption history. JSON output (`{score, reasoning}`) so downstream code can parse.

```
{
  "score": 78,
  "reasoning": "Active severe T-storm
warning over TN + two MEM hub
diversions in past 24h significantly
elevate likelihood of further sort."
}
```

06

Action layer — write-back with HITL approval

The difference between a report and an operational system. The Composer proposes 2-4 actions per brief; each lands in the UI's Action Queue with Approve / Reject buttons. On approve, the action commits (in production: Jira, Slack, carrier comms; here: Delta + JSONL audit trail).

 **propose_action**

generic write-back

Any imperative-verb-led action with severity, owner_role, impact estimate, related_objects. The canonical action proposal — Composer calls this 2-4x per brief.

 **file_disruption_ticket**

Jira-shape draft

Drafts a disruption-tracking ticket with affected carriers, FCs, lanes, and severity. Returns the draft (not yet sent — awaiting approval).

 **draft_carrier_email**

outbound comms draft

Drafts an email to a named carrier (FedEx/UPS/etc.) with subject, body markdown, impacted-shipment count. Lands in the Action Queue.

 **log_decision**

decisions_log audit trail

Appends a brief-level decision (headline, severity, themes, rationale) to the decisions log — both an in-memory ring buffer and a JSONL file.

Action persistence

Every proposed action is double-written: (a) into an in-process `deque(maxlen=50)` the UI reads via `GET /actions/queue` , and (b) into `/tmp/ops_brief/actions.jsonl` for offline audit. Approve / Reject events append a new row with the operator's identity and timestamp.

07

Vector Search recall (Historian)

Stage 4 — Memory + MCP

14 nights of synthetic past briefs in a Delta table, embedded with `databricks-gte-large-en` , indexed in Vector Search. The Historian semantically queries past briefs and surfaces recurring themes.

The setup

- ✓ Delta table `briefs_archive` · 14 rows (May 14–27 2026) · CDF enabled
- ✓ VS endpoint `ops_brief_vs` · STORAGE_OPTIMIZED · ONLINE
- ✓ VS index `briefs_archive_index` · Delta-Sync · TRIGGERED · embedding on `body_markdown`
- ✓ Agent tool `recall_similar_briefs(query, days_back)` wired into Historian
- ✓ Delta-SELECT fallback if VS unavailable

Live smoke test output

```
query: "Memphis weather impact"

● LIVE — Databricks Vector Search
hits: 3 (window: last 14 days)

1. 2026-05-27 sev=69.0 similarity=0.620
   Memphis severe weather AGAIN — recurring
   storm pattern this month
   themes=[MEM weather, FedEx hub risk, recurring]

2. 2026-05-16 sev=64.0 similarity=0.607
   themes=[MEM weather, FedEx delayed sort]

3. 2026-05-15 sev=58.0 similarity=0.577
   themes=[MEM weather, FedEx ground hold]
```

Cross-checked with *"Port of LA labor strike"* — correctly returned briefs from 5/18, 5/19, 5/20 (the consecutive labor-unrest cluster). Semantic discrimination is real, not just keyword overlap.

08

Operational Console UI

Not a chatbot. A 4-pane operational console — Palantir Foundry style, dense, light-theme-only, Nunito Sans + ShipBob design tokens. Single self-contained HTML page (~823 lines), no external assets except Google Fonts and marked.js for markdown rendering.

The 4 panes

- ✓ Left rail — Signals. 17 live tiles (status dot · count · last-updated) refreshed every 60s via `/signals/latest`
- ✓ Center — Brief. Rendered markdown with clickable object chips `[OBJ:FC-MEM]` and provenance chips `[CITE:query-id]`
- ✓ Right rail — Actions. Approve/Reject buttons. Card flips to ✓ APPROVED on commit.
- ✓ Bottom — Timeline. Live supervisor → subagent → tool call transitions, color-coded by actor

Investigation drill-down

- Click any object chip in the brief — `[OBJ:FC-MEM]` , `[OBJ:CARRIER-FEDEX]` , etc. — a slide-in panel pulls live properties from the SQL warehouse:
- ✓ Properties: capacity, in-flight, at-risk, region
 - ✓ Links: top carriers, active disruptions, related shipments
 - ✓ Chained navigation: click any linked chip → new card
 - ✓ Foundry-style Object Cards over Databricks Apps

Severity gauge + JSON sidecar

The Composer emits a trailing ````json {severity, pending_actions}```` block. The UI parses it, strips it from the displayed brief, animates the severity gauge from 0→7.2, and populates the Action Queue. Green <4, amber 4-7, red ≥7.

Endpoint surface

METHOD	PATH	PURPOSE
POST	<code>/invocations</code>	Agent run (MLflow ResponsesAgent contract)
GET	<code>/signals/latest</code>	17-source fan-out, cached 60s

METHOD	PATH	PURPOSE
GET	/actions/queue	Pending action proposals
POST	/actions/approve	HITL approve
POST	/actions/reject	HITL reject
GET	/decisions	Recent decision-log entries
GET	/objects/lookup?id=...	Foundry-style object card lookup
GET	/debug/health	Env, tools, error count
GET	/debug/logs	Last 500 in-process log lines
GET	/debug/cost	Per-request token + tool-call telemetry

09

MLflow Agent Evaluation

Stage 3 — MLflow

20-item golden dataset · 3 LLM-as-judge scorers · 911-line harness. Results land as logged tables + metrics in MLflow experiment 4469475842996134.

Scorers

groundedness

Every numeric claim and named entity traces to source. Hallucinated FCs/carriers/\$ = low score.

action_quality

Are proposed actions specific (verb + object + owner) and proportional to severity?

subagent_routing

Did the supervisor invoke the expected subagent set? Jaccard + LLM-judge hybrid.

Run results

RUN	ITEMS	GROUNDEDNESS	ACTION_QUALITY	SUBAGENT_ROUTING	AVG LATENCY	MLFLOW
Smoke (Run 1)	2	0.900	0.875	0.812	67.6s	ddd2ef0b
Run 2	5	0.600	0.900	0.685	111.8s	8804a5d5

⚠ 5-item **groundedness dip**: one item ("which carrier has the most at-risk shipments?") hit a schema gap — per-carrier breakdown isn't yet a Metric View dimension. The agent caveated transparently; judge gave partial credit. Documented improvement in [EVAL_RESULTS.md](#).

10

Databricks AI Maturity Ladder — all 4 stages

Databricks' official enablement slide teaches a 4-stage maturity ladder. Most projects hit one or two. This project visibly touches every stage.

STAGE	STACK NAME	WHERE IT LIVES IN THIS PROJECT
Stage 1	Text-to-SQL & Coding · Genie + Genie Code + AI/BI Genie	Managed Genie MCP server is the DataAnalyst's only data path. Genie Code customization workflows in GENIE_CODE_CUSTOMIZATION.md (explicit scoring criterion).
Stage 2	AI Functions · AI Gateway + ai_query/ai_classify	3 governed UC SQL UDFs (classify_disruption_severity , summarize_signal_pulse , carrier_risk_score) powered by ai_query / ai_classify , exposed to the agent via UCFunctionToolkit .
Stage 3	Build-Your-Own Agent · Agent Framework + Agents on Apps + MLflow	The whole multi-agent supervisor on a deployed Databricks App. MLflow 3 autolog + Agent Evaluation harness with 3 LLM-as-judge scorers.
Stage 4	Agent Memory & Tools · Lakebase + MCP	Managed Genie MCP (data tools). Vector Search briefs archive for semantic recall (Historian). Lakebase wired (currently disabled — SP grant pending; falls back to InMemorySaver).

11 Judging rubric — how this project scores

Track 1 Section 6 official rubric (verbatim weights). Below: each criterion mapped to what we built. Technical Excellence + Innovation + Business Impact = 75% of total; Demo is only 10% — *depth wins over polish*.

1. Technical Excellence

25%

WHAT SCORES	Effective use of Genie Code as agentic coding tool · Code quality/scalability · UC, Delta Lake, UC Metric Views, AI/BI Dashboards · Data security, governance, cost · <i>Instructions to customize Genie Code</i> · Adding MCP servers · Production readiness
WHAT WE SHIPPED	• GENIE CODE AS PRIMARY WORKFLOW — `GENIE_CODE_CUSTOMIZATION.md` covers 4 named extension patterns (add signal / add UC AI Function / add subagent / add action) • 3 MCP INTEGRATIONS — Managed Genie MCP, langchain_mcp_adapters at request time, UC Function MCP-equivalent via toolkit • UC METRIC VIEWS over Gold tables (3 measures: In Flight Count, At Risk Count, Revenue At Risk USD), Genie space scoped to MVs only • PRODUCTION-READINESS: graceful Lakebase fallback (no 30s hang on unreachable instance); 17 synthetic-fallback signal tools; MLflow eval harness with 3 scorers; cost telemetry middleware; debug endpoints (/health, /logs, /cost); MERGE-idempotent setup scripts

2. Innovation & Creativity

25%

WHAT SCORES	Originality of use case · Novel application of Genie Code · Creative integration of multiple Databricks features
WHAT WE SHIPPED	• PLAN-DRIVEN SUPERVISOR (not ReAct-with-N-tools) — Pydantic structured-output planning is visible in MLflow traces • 17 REAL-TIME EXTERNAL SIGNALS FUSED WITH INTERNAL GENIE DATA — most multi-agent projects either go internal-only or external-only; we do both with parallel fan-out • FOUNDRY-STYLE OBJECT CARDS INSIDE A DATABRICKS APP — clickable [OBJ:FC-MEM] chips drill into live warehouse data, chain to related entities • ACTION LAYER WITH HITL APPROVAL — write-back actions (file ticket, draft email, log decision) queue for human approval • NIGHTLY DATABRICKS JOB (PAUSED) — proactive mode, "vertical workflow not chatbot" pattern • COST TELEMETRY — per-request token + subagent-call counters surfaced at /debug/cost

3. Business Impact & Relevance

25%

WHAT SCORES	Clarity of business problem · Potential customer value · Applicability to real-world enterprise scenarios
WHAT WE SHIPPED	• REAL, UNIVERSAL PROBLEM — every supply-chain operator has the "what hit overnight?" question. Every DNB customer with physical shipments cares. • QUANTIFIED IMPACT — 4,592 in-flight tonight · 216 at-risk · \$192,704 revenue at risk · 4 actions proposed and queued for human approval • DAILY 30-MIN OPS STANDUP COLLAPSED TO 30-SECOND SCAN — that's the headline ROI • APPLIES BEYOND LOGISTICS — same architecture works for finance ops (overnight market events), security ops (alert triage), platform ops (incident sweep). Domain-agnostic shape, domain-specific signals.

4. Data & App Quality

15%

WHAT SCORES	Functionality of Dashboards/Apps · UX for non-technical stakeholders · Quality of insights
WHAT WE SHIPPED	• 4-PANE OPERATIONAL CONSOLE — signals · brief · actions · timeline. Light-theme. Dense, not chatbot-generic. • 17 LIVE SIGNAL TILES REFRESHING EVERY 60S with status dots (live/synthetic/stale) — instant situational awareness without reading the brief • SEVERITY GAUGE ANIMATES 0-10 with color thresholds (green <4, amber 4-7, red ≥7) • OBJECT CHIPS → DRILL-DOWN PANEL with chained navigation across FC → carrier → shipment → customer • DEBUG PANEL with Health / Logs / Errors / Copy-for-Claude buttons — judges can audit live

5. Demo

10%

WHAT SCORES	Clarity and storytelling · Communication to both technical and business audiences
-------------	---

- **3-MIN RECORDING PLAN** in `DEMO\_SCRIPT.md` — timestamped, voice-over cues, architecture diagram on screen, MLflow cut, all 5 chat scenarios pre-canned
- **SUBMISSION SUMMARY** `SUBMISSION.md` — every Rule §4.3 deliverable explicitly addressed
- **THIS OVERVIEW PAGE** — judges can pre-watch the architecture before the video

- ✔ **Working URL** — <https://ops-brief-apatel-7474650842988229.aws.databricksapps.com> · App in RUNNING state
- 🕒 **Code link** — push to a GitHub repo (the form needs a code URL)
- 🕒 **Screen recording demo** — ~3 min, recorded on Lighthouse environment, per `DEMO\_SCRIPT.md`
- ✔ **Project description** — `SUBMISSION.md` + `README\_OPS\_BRIEF.md`
- ✔ **Lighthouse URL** — workspace `dnb-hackathon-west-2`
- ✔ **Original work** — all artifacts timestamped May 22+ 2026 (within registration window)
- ✔ **Genie Code customization doc** — `GENIE\_CODE\_CUSTOMIZATION.md`
- ✔ **MLflow eval results** — `EVAL\_RESULTS.md` + 2 logged MLflow runs
- ⚠ **Submission deadline:** Friday May 29 2026, 6:00 PM PT

```
ops-brief-project/      # synthetic data + Genie + UC Fns + nightly job
├─ data/
│   └─ schema.sql      # 9 Delta tables (~30k synthetic shipments)
├─ scripts/
│   ├── setup_data.py  # synth data + Metric Views (~3 min)
│   ├── create_genie_space.py  # Genie space + 5 verified queries
│   ├── setup_uc_functions.py  # 3 UC AI Functions (ai_classify / ai_query)
│   ├── setup_briefs_archive.py  # 14-day briefs archive + Vector Search index
│   └─ register_nightly_job.py  # Databricks Job (23:00 CT, PAUSED)

databricks-agent-on-apps-genie-mcp/  # the multi-agent app
├─ agent_server/
│   ├── supervisor.py   (700+ LoC)  # LangGraph supervisor + 7 nodes + Plan schema
│   ├── agent.py        (240 LoC)   # MLflow ResponsesAgent wiring
│   ├── external_tools.py  (244 LoC) # 4 original signal tools
│   ├── external_tools_extra.py (517) # 5 more signal tools
│   ├── external_tools_v3.py (875 LoC) # 8 more signal tools (FAA/CBP/HN/EIA/...)
│   ├── uc_function_tools.py (56 LoC) # UCFunctionToolkit binding
│   ├── brief_tools.py   (92 LoC)   # compose_brief
│   ├── action_tools.py  (326 LoC)   # 4 write-back action tools + ring buffer
│   ├── recall_tools.py  (300+ LoC)  # Vector Search recall (Historian)
│   ├── investigation_tools.py (240+) # Object lookup (drill-down)
│   ├── cost_telemetry.py (175 LoC)  # per-request token + tool counters
│   ├── endpoints.py     (220 LoC)   # /signals, /actions, /objects, /decisions
│   ├── chat_ui.py       (~900 LoC)  # 4-pane operational console (single page)
│   ├── debug_view.py    (89 LoC)   # /debug/health, /debug/logs
│   ├── start_server.py  (110+ LoC)  # FastAPI assembly + lifespan
│   ├── utils_memory.py  (254 LoC)   # Lakebase wiring (degraded fallback)
│   └─ utils.py          (346 LoC)   # MLflow + MCP client
├─ scripts/
│   └─ evaluate_agent.py  (911 LoC)  # 20-item golden + 3 LLM-as-judge scorers
├─ databricks.yml        # bundle config (app, resources)
├─ DEMO_SCRIPT.md        # 3-min recording plan
├─ GENIE_CODE_CUSTOMIZATION.md  # rubric scoring criterion
└─ SUBMISSION.md         # formal submission summary
```



```
└─ EVAL_RESULTS.md          # MLflow eval scores + improvement roadmap
└─ PROJECT_OVERVIEW.html    # this page
```

Built May 22–29 2026 · 17 signal sources · 5 specialist subagents · 3 UC AI Functions · 4 write-back actions · 14-day Vector Search recall · MLflow 3 traced · Genie MCP throughout
· Light-theme operational console · ShipBob brand tokens

This is a hackathon submission. No real company data anywhere. All synthetic. All open-source signal feeds.